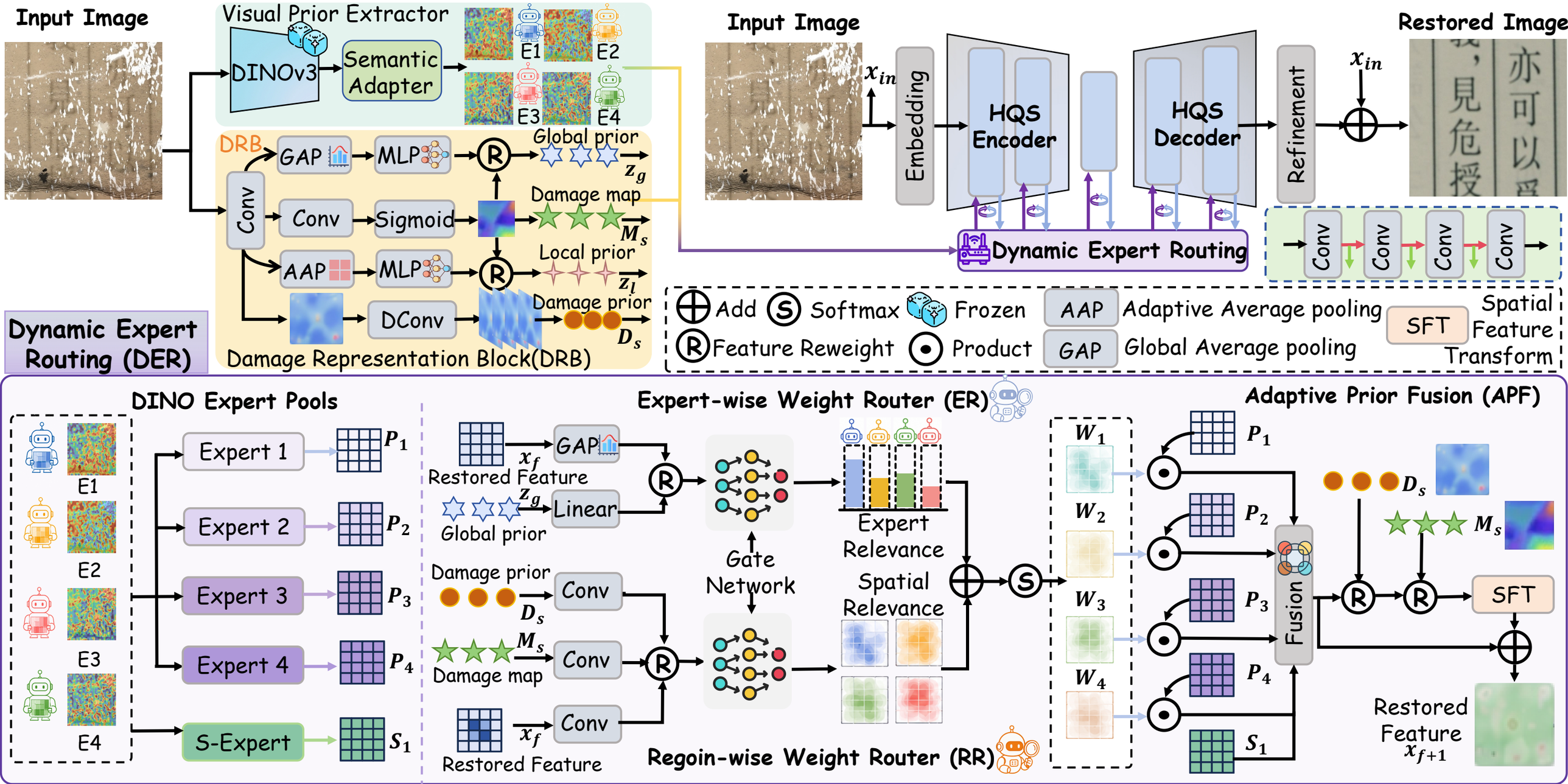

Input Image
Visual Prior Extractor
DINOv3
Semantic Adapter
E1
E2
E3
E4
DRB
GAP
MLP
Global prior
z_g
Conv
Conv
Sigmoid
Damage map
M_s
AAP
MLP
Local prior
z_l
DConv
Damage prior
D_s
Damage Representation Block(DRB)
Dynamic Expert Routing (DER)
Input Image
x_{in}
Embedding
HQS Encoder
HQS Decoder
Refinement
Restored Image
Dynamic Expert Routing
Conv
Conv
Conv
Conv
Add
Softmax
Frozen
AAP
Adaptive Average pooling
SFT
Spatial Feature Transform
Feature Reweight
Product
GAP
Global Average pooling
DINO Expert Pools
E1
E2
E3
E4
Expert 1
Expert 2
Expert 3
Expert 4
S-Expert
P_1
P_2
P_3
P_4
S_1
Expert-wise Weight Router (ER)
Restored Feature
x_f
GAP
Global prior
z_g
Linear
Gate Network
Expert Relevance
Damage prior
D_s
Conv
Damage map
M_s
Conv
Spatial Relevance
Restored Feature
x_f
Conv
Regoin-wise Weight Router (RR)
Adaptive Prior Fusion (APF)
W_1
W_2
W_3
W_4
P_1
P_2
P_3
P_4
S_1
Fusion
D_s
M_s
SFT
Restored Feature
x_{f+1}